

CRPA Hardware Architecture

Technical Report - Interview Assignment

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Stage 0 - Background Summary

CRPA operating principle

A Controlled Reception Pattern Antenna (CRPA) uses multiple spatially separated antenna elements and complex channel weights to shape the receive pattern. A signal arriving from a given direction reaches the elements with deterministic phase differences. By applying complex weights and summing the channels, the array can produce destructive interference in the jammer direction while preserving useful GNSS directions. The adaptive weight-generation algorithm itself is assumed to be available in the FPGA, as required by the assignment; the hardware must preserve coherent amplitude and phase so that the algorithm can form deep and stable nulls.

$$y(t) = \sum_n w_n x_n(t)$$

- Spatial suppression depends on relative amplitude and phase accuracy across channels.
- The channel weights are adjusted to place nulls toward interference directions while maintaining gain toward useful GNSS signals.
- Clock synchronization, LO coherence and calibration directly limit the achievable null depth.

Receiver architectures

Architecture	Signal path	Main advantage	Main limitation
Superheterodyne	RF → IF → ADC	Strong analog selectivity and blocker handling	More mixers/LOs/filters; higher part count and SWaP
Zero-IF	RF → I/Q baseband → ADC	Compact integrated RFIC; direct complex I/Q	DC offset, 1/f noise and I/Q imbalance must be calibrated
Direct RF sampling	RF → ADC → DDC	Minimal analog conversion and flexible digital processing	Very high ADC rate, clock-jitter sensitivity and data/power burden

GNSS signal structure

Open-service GNSS signals are spread-spectrum RF signals composed of a carrier, a pseudo-random spreading code and navigation data. The spreading code expands the signal bandwidth and enables code-domain acquisition and tracking at very low received power levels. GNSS signals are typically received at power levels on the order of -130 dBm, depending on the signal and reception conditions [2]–[5]. GNSS receivers recover these weak signals through spreading-code correlation and processing gain.

Signal property	Hardware relevance
Carrier frequency	Defines RF tuning, preselector bands and LO plan.
Spreading-code rate / modulation	Sets the useful signal bandwidth and therefore required analog/digital bandwidth.
Very low received power	Drives low-noise front-end design and the need to avoid unnecessary pre-LNA loss.
Simultaneous multi-GNSS operation	Requires coverage of L1, L2 and L5 families without losing coherent phase relationships.

Signals used in this assignment

Group	Open signals	Center-frequency region
L1	GPS L1 C/A, Galileo E1 OS, BeiDou B1I, GLONASS G1 C/A	1559-1606 MHz overall
L2	GPS L2C, Galileo E5b, BeiDou B2I, GLONASS G2 C/A	1197-1249 MHz overall
L5	GPS L5, Galileo E5a, BeiDou B2a	1166-1187 MHz overall

Design implication

The front end must preserve weak GNSS signals while tolerating in-band jammers that can be tens of decibels stronger than the integrated thermal-noise floor. This leads to three hardware priorities used throughout the report: low front-end noise figure, sufficient blocker/dynamic-range margin, and coherent synchronization/calibration across all antenna channels.

1. Signal Analysis and Frequency Plan

The center frequencies are taken directly from the assignment brief. Main-lobe bandwidth values are based on the corresponding open-service signal definitions; where modulation definitions differ from transmitter occupied bandwidth, the value used here is explicitly treated as the processing/main-lobe bandwidth for the preliminary front-end plan [1]-[6].

Group	Signal	Center frequency (MHz)	Main-lobe BW (MHz)
L1	GPS L1 C/A	1575.420	2.046
L1	Galileo E1 OS	1575.420	14.322
L1	BeiDou B1I	1561.098	4.092
L1	GLONASS G1 C/A	$1602 + k \cdot 0.5625$, $k=-7 \dots +6$	1.022
L2	GPS L2C	1227.600	2.046
L2	Galileo E5b	1207.140	20.460
L2	BeiDou B2I	1207.140	4.092
L2	GLONASS G2 C/A	$1246 + k \cdot 0.4375$	1.022
L5	GPS L5 / Galileo E5a / BeiDou B2a	1176.450	20.460

Band coverage and grouping decision

Band	Required RF span	Instantaneous span
L1	1559.052-1605.886 MHz	46.834 MHz
L2	1196.910-1249.136 MHz	52.226 MHz
L5	1166.220-1186.680 MHz	20.460 MHz
Selected L2+L5 chain	1166.220-1249.136 MHz	82.916 MHz

L2 and L5 are close enough to share one receive chain, while placing L1 in the same analog path would expand the instantaneous span to 439.666 MHz.

Therefore the selected architecture uses two RF paths per antenna element: one L1 path and one shared L2+L5 path. This reduces hardware compared with three independent chains without imposing the very wide bandwidth of a single-chain solution.

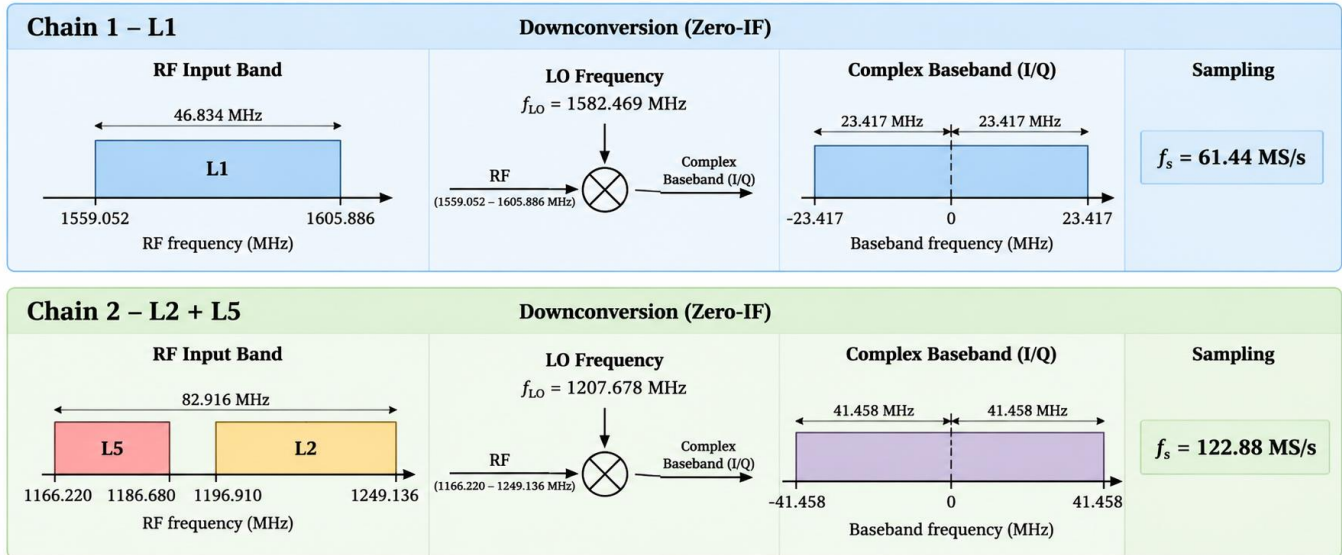


Figure 1. Preliminary two-chain frequency plan. Chain 1 covers the L1 band (1559.052–1605.886 MHz) with LO = 1582.469 MHz and $f_s = 61.44$ MS/s. Chain 2 covers the L2 and L5 bands (1166.220–1186.680 MHz and 1196.910–1249.136 MHz) with LO = 1207.678 MHz and $f_s = 122.88$ MS/s. Zero-IF downconversion is used for both chains, resulting in complex baseband spectra of ± 23.417 MHz (L1) and ± 41.458 MHz (L2+L5).

2. System Architecture

2.1 Architecture trade-off and selection

Criterion	Zero-IF RFIC	Direct RF sampling
Performance	Adequate bandwidth; integrated gain/attenuation and complex I/Q	Excellent flexibility; strongest ADC/clock requirements
Component count	Low: RFIC integrates mixer, LO/PLL, ADC and digital interface	Analog chain is simple but converter/clocking subsystem is demanding
Power	Moderate	Typically higher for multi-GSPS ADCs and data transport
Mass	Low/moderate	Potentially higher due to converter, clock and thermal hardware
Cost	Moderate/high RFIC cost	High-performance ADC/clocking cost
Design ease	Integrated calibration and JESD support	Higher clock-jitter, aliasing and digital-interface complexity

Zero-IF is selected because the required instantaneous bandwidths (46.834 MHz and 82.916 MHz) are comfortably below the 200 MHz ADRV9026 receive bandwidth, while the integrated four-channel transceiver significantly reduces analog part count and supports coherent multi-device synchronization [10]. Direct RF sampling remains technically attractive, but its converter rate, data throughput, clock-jitter and power requirements are less favorable for the 50 W / 2 kg UAV DACU budget.

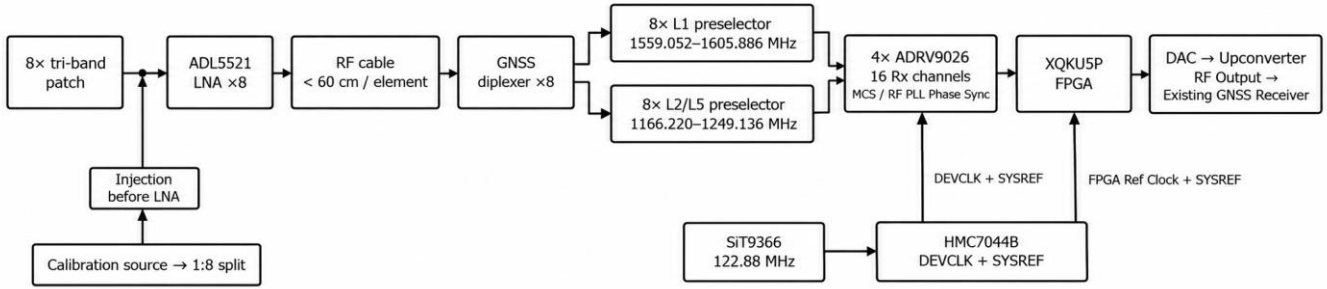


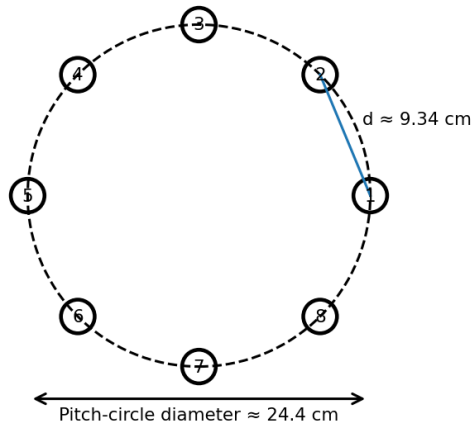
Figure 2. Selected 8-element hardware architecture. Four ADRV9026 devices provide 16 coherent receive channels (8 elements $\times$ 2 RF paths).

Component placement

The ADL5521 LNA is placed at the antenna side, immediately after each tri-band patch. The <60 cm RF cable then carries the amplified signal to the DACU, minimizing the contribution of cable loss to the cascaded noise figure. Inside the DACU, 8 $\times$ GNSS diplexers separate the L1 and L2/L5 paths, followed by 8 $\times$ L1 and 8 $\times$ L2/L5 preselectors before four ADRV9026 devices providing 16 coherent receive channels. Cable and PCB losses are neglected in the first-order cascade calculation and are listed explicitly as an assumption.

2.2 Array geometry and scalability

An 8-element uniform circular array (UCA) is used for the reference configuration. Spacing is set to approximately $\lambda/2$ at the highest required L1 frequency (1605.886 MHz), which gives $\lambda \approx 18.68$ cm and adjacent spacing $d \approx 9.34$ cm. For an 8-element UCA this corresponds to a pitch-circle diameter of approximately 24.4 cm.



Geometry check

- $f_{\max} = 1605.886$ MHz
- $\lambda_{\min} \approx 18.68$ cm
- $d \approx \lambda/2 \approx 9.34$ cm
- Pitch-circle diameter ≈ 24.4 cm
- 35 cm outer-diameter limit leaves ≈ 5.3 cm radial allowance for the patch footprint.

The passive patch dimensions are not specified by the assignment. Final 35 cm mechanical compliance must therefore be verified against the selected antenna element footprint.

First-order scaling table

Elements	RF paths	ADRV9026 count	Payload data rate	Power estimate	Mass estimate	Budget
4	8	2	23.59 Gb/s	~20 W	~0.8 kg	Within
8	16	4	47.19 Gb/s	~40 W	~1.6 kg	Within
12	24	6	70.78 Gb/s	~60 W	~2.4 kg	Exceeds
16	32	8	94.37 Gb/s	~80 W	~3.2 kg	Exceeds

Scaling power and mass are first-order linear extrapolations from the 8-element reference budget. They are intended to identify where the fixed DACU limits become infeasible, not to replace a detailed layout/thermal model.

3. RF Front End and Cascade Analysis

3.1 Interference levels and suppression strategy

Free-space path loss is calculated as

$FSPL(dB) = 32.44 + 20\log_{10}(d_{km}) + 20\log_{10}(f_{MHz})$. The received interference power at each antenna-element output is $EIRP - FSPL + G_{element}$, using the direction-dependent element gains given in the assignment [1].

Source	FSPL	Element output power	Classification	Primary suppression
J1: 1575.42 MHz CW	110.37 dB	-75.37 dBm	In-band L1	Spatial null
J2: 1176.45 MHz / 20 MHz	111.91 dB	-73.91 dBm	In-band L5	Spatial null
J3: 1486 MHz / 20 MHz	107.92 dB	-65.92 dBm	Out-of-band	Preselector filter

Reference noise and suppression targets

For a 290 K first-order reference, integrated thermal-noise power is $N = -174 + 10\log_{10}(B)$ dBm. Using the full selected RF spans gives -97.3 dBm for L1 (46.834 MHz) and -94.8 dBm for L2+L5 (82.916 MHz). Representative open-signal powers are approximately -128.5 dBm (GPS L1 C/A), -130 dBm (GPS L2C), and -127.9 dBm (GPS L5) [2],[3]. At the antenna-element output, J1 is approximately 53.1 dB stronger than the GPS L1 C/A reference, J2 is approximately 54.0 dB stronger than the GPS L5 reference, and J3 is approximately 62.6 dB above the L1 reference level. J3 is out of band and is therefore handled primarily by the preselector filter.

Source	Comparison	Required suppression criterion
J1	-75.4 vs -97.3 dBm noise floor	≈21.9 dB spatial suppression
J2	-73.9 vs -94.8 dBm noise floor	≈20.9 dB spatial suppression
J3	-65.9 vs -97.3 dBm L1 noise floor	≈31.4 dB; design target ≥32 dB filter rejection

The preliminary criterion is intentionally conservative and simple: reduce each dominant interferer to approximately the corresponding integrated thermal-noise floor before receiver NF. It is used to size filtering and calibration, not as a final tracking-performance model.

Front-end component selection

Function	Selected part	Key design reason
LNA	ADL5521ACPZ-R7	400-4000 MHz coverage, ~1 dB NF class, high P1dB/OIP3 [7]
GNSS diplexer	1224DP15A1572001E	1164-1285 / 1539-1606 MHz, 1 dB max insertion loss [8]
L1 preselector	AM1600S1059	1550-1650 MHz passband; ≥45 dB rejection over 1200-1500 MHz [9]
L2/L5 preselector	AFII-SW-0065	1166–1300 MHz coverage; low L5/L2 loss; ≥35 dB rejection at 1486 MHz [11]

The order LNA → diplexer → preselector is selected to protect the overall noise figure: placing the LNA first reduces the contribution of the following 1–4 dB passive losses to the cascaded NF. The trade-off is that the LNA must tolerate the incoming blockers; its linearity is therefore checked together with the RFIC headroom.

3.2 Cascade, ADC headroom and dynamic range

Stage	L1 gain / NF	L2+L5 gain / NF	Linearity note
ADL5521	+13 dB / ~1 dB	+15 dB / ~1 dB	OP1dB ≈ +21.8 dBm class [7]
GNSS diplexer	-1.0 dB / 1.0 dB	-1.0 dB / 1.0 dB	Passive
Preselector	-4.0 dB / 4.0 dB	-1.2 dB / 1.2 dB	Passive
ADRV9026	Variable gain / ~11.5 dB NF	Variable gain / ~11.5 dB NF	IIP3 > +15 dBm class; Rx attenuation available [10]

$$F_{total} = F_1 + (F_2 - 1)/G_1 + (F_3 - 1)/(G_1 G_2) + (F_4 - 1)/(G_1 G_2 G_3)$$

Using the design gains/losses above, the cascaded noise figure is approximately 5.4 dB for L1 and 2.9 dB for L2+L5. The L1 result is dominated by the 4 dB preselector loss after a relatively modest first-stage gain; the lower-band chain benefits from lower filter loss and slightly higher LNA gain.

Interference power at the RFIC/ADC input

Path	Total receiver-input power	ADRV9026 reference	Full-scale headroom
L1	≈ -67.0 dBm	P_HIGH ≈ -12.7 dBm at -2 dBFS	≈56.3 dB
L2+L5	≈ -60.8 dBm	P_HIGH ≈ -12.7 dBm at -2 dBFS	≈50.1 dB

The total powers are obtained by linear power summation of the residual interferers and noise at the ADRV9026 input. Because P_HIGH corresponds to -2 dBFS, the equivalent 0 dBFS input reference is approximately -10.7 dBm; the resulting headroom remains above 50 dB in both paths. ADRV9026 uses 16-bit Rx ADCs and Analog Devices specifies approximately 81 dBc Rx SFDR, leaving about 25 dB SFDR margin relative to the ~56 dB worst-case input span [10],[12].

AGC and additional-interferer assessment (optional)

- Keep the external LNA gain fixed; apply common ADRV9026 Rx attenuation/AGC settings across channels so relative amplitude and phase are preserved.
- A 1240 MHz, 1 μ s / 1 kHz pulsed radar lies in the L2+L5 path. A sufficiently high peak can temporarily saturate the LNA/RFIC/ADC; clipped samples cannot be recovered by digital nulling.
- A 1621 MHz satphone signal lies inside the wide L1 preselector passband but above the 1539-1606 MHz diplexer L1 passband; its rejection therefore depends mainly on the diplexer skirt. EIRP and range are not specified, so this check remains qualitative.

Detailed cascade equations, power summation and datasheet fields should remain visible in cascade.xlsx, as required by the assignment.

4. Synchronization, Calibration and Digital Architecture

4.1 Clock/LO synchronization and calibration

All receive channels must remain phase coherent. A SiT9366 122.88 MHz master reference feeds an HMC7044B clock/SYSREF distribution device. HMC7044B distributes deterministic device clocks and SYSREF to the ADRV9026 devices and FPGA. The four ADRV9026 RF PLLs are programmed to the same LO frequencies and synchronized using multi-chip synchronization / RF PLL phase synchronization. SYSREF aligns converter/JESD timing; it does not replace the continuous device clock [10],[13],[14].

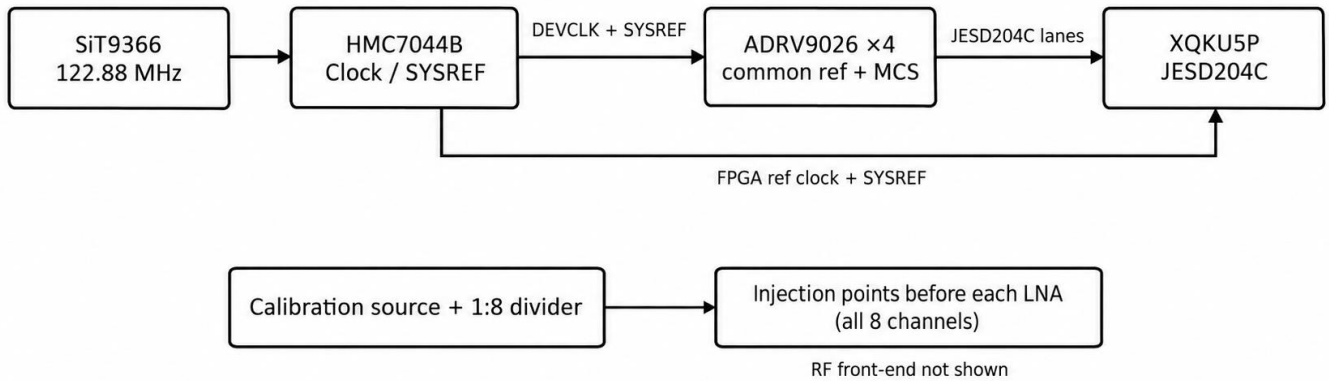


Figure 3. Reference-clock, SYSREF and calibration concept. A common calibration signal is injected before each LNA to measure complete channel mismatch.

Calibration accuracy versus null depth

$$\text{Null-depth limit (dB)} \approx 10 \log_{10}(\epsilon a^2 + \epsilon \phi^2), \quad \epsilon a = 10^{(\Delta A/20)} - 1$$

Amplitude error ΔA	Phase error	Null-depth limit
0.5 dB	5°	-19.5 dB
0.2 dB	2°	-27.5 dB
0.1 dB	1°	-33.6 dB
0.05 dB	0.5°	-39.6 dB

Stage 3 requires approximately 21.9 dB and 20.9 dB spatial suppression for J1 and J2. A calibration tolerance of ≤ 0.2 dB amplitude error and $\leq 2^\circ$ phase error is therefore selected, corresponding to an approximate -27.5 dB residual limit and providing several decibels of calibration margin. The common calibration tone is split 1-to-8 and injected before each LNA so that the measured correction includes the LNA, diplexer, filter and RFIC path.

4.2 Data interface, FPGA and RF output

Each antenna element produces one complex L1 stream at 61.44 MS/s and one complex L2+L5 stream at 122.88 MS/s. With 16-bit I and 16-bit Q, the aggregate payload rate for eight elements is:

$$R_{\text{total}} = 8 \times (61.44 + 122.88) \text{ MS/s} \times 32 \text{ bit/sample} = 47.19 \text{ Gb/s}$$

JESD204C is selected because ADRV9026 natively supports JESD204B/C and JESD204C provides lower 64b/66b coding overhead than JESD204B. Including

coding overhead, the aggregate serial payload is approximately 48.7 Gb/s before implementation-specific framing. The four RFICs use separate serial links into the FPGA; streams are combined only inside the FPGA.

FPGA selection

Device	High-speed I/O	DSP / memory	Selection rationale
AMD Kintex UltraScale+ XQKU5P	16 GTY transceivers, up to 28.2 Gb/s	1824 DSP, ~34 Mb memory	Sufficient JESD204C lane capacity, compact defense-grade device, suitable temperature range [15]

The 16 GTY transceivers provide sufficient aggregate serial bandwidth for the four ADRV9026 JESD204C links under the selected interface configuration. FPGA power is design-dependent; a 10 W preliminary allocation is therefore used only for SWaP budgeting and must be replaced by AMD power-estimator / post-implementation results.

RF output and thermal phase stability

The processed beamformed signal is converted back to RF through a single output chain: FPGA → DAC → upconverter → RF output → existing GNSS receiver. Receiver-specific output level, antenna-port DC bias and DAC-spur requirements are not numerically defined in the assignment and are therefore not fixed in this preliminary design. Temperature variation can change electrical delay in cables, PCB traces and RF devices, creating channel-to-channel phase drift and reducing null depth. Periodic injected-tone calibration is proposed to update the phase-correction coefficients over temperature.

5. BOM, SWaP-C and Supply Risk

5.1 Critical-component BOM

Component	Part number	Quantity	Approx. unit price	Status
LNA	ADL5521ACPZ-R7	8	~\$5.4	Active / production
L1 filter	AM1600S1059	8	RFQ	In stock / current product
L2/L5 filter	AFII-SW-0065	8	~\$2.6	Active / new product
RFIC / ADC	ADRV9026BBCZ	4	~\$1,409	Production
FPGA	XQKU5P-1FFRB676I	1	~\$4,000 budgetary	Current XQ family
Reference oscillator	SiT9366 (122.88 MHz)	1	~\$25	Active

Prices are approximate single-unit/budgetary web values used only to identify cost drivers; actual procurement pricing requires distributor/manufacturer quotation.

5.2 SWaP-C budget

POWER BUDGET

- 4× ADRV9026: 16.76 W
- 8× ADL5521: 2.40 W
- XQKU5P FPGA: ~10.0 W
- HMC7044B: 1.93 W
- SiT9366: 0.26 W
- DAC + upconverter reserve: ~2.0 W
- Electronics subtotal: ~33.4 W
- With 20% design margin: ~40 W
- Requirement: ≤50 W → ~10 W margin

MASS BUDGET

- PCB + electronics: ~0.35 kg
- Aluminum enclosure + baseplate: ~0.75 kg
- Connectors/shielding/wiring: ~0.20 kg
- Mechanical hardware: ~0.10 kg
- Engineering reserve: ~0.20 kg
- Estimated total: ~1.60 kg
- Requirement: ≤2.0 kg → ~0.40 kg margin

Main cost drivers: 4× ADRV9026 RFICs and the XQKU5P FPGA, because their unit costs are orders of magnitude above the remaining RF components.

5.3 Supply risk and alternatives

Critical item	Risk	Candidate alternative
ADRV9026	High-cost specialized RFIC; single-vendor dependency	TI AFE7950 (system-level alternative; direct-RF architecture requires requalification)
XQKU5P	High-cost defense-grade FPGA; vendor dependency	XCKU5P-2FFVB676I (similar resource class; not defense-grade XQ)
AM1600S1059	Specialty L1 filter with limited second-source options	EMA4F1580P80G30B candidate; cascade/environmental requalification required

Assumptions and References

Assumptions table

ID	Assumption
A1	Main-lobe/reference bandwidth values are used for preliminary analog coverage; exact occupied-spectrum definitions vary by modulation.
A2	Final sample rates are 61.44 MS/s (L1) and 122.88 MS/s (L2+L5).
A3	Cable/PCB losses are neglected in the first-order Friis calculation; LNA is placed antenna-side so cable loss occurs after first gain.
A4	Spatial/filter suppression targets use residual-interference $\leq$ integrated 290 K thermal-noise floor as a preliminary criterion.
A5	ADRV9026 11.5 dB NF and -12.7 dBm P_HIGH are conservative 1.8 GHz reference values used for both paths.
A6	LNA gains of +13 dB (L1) and +15 dB (L2+L5) are representative cascade-model values, not guaranteed across all matching/layout implementations.
A7	FPGA power allocation (~10 W) and DAC/upconverter reserve (~2 W) are preliminary; final values require implementation/component selection.
A8	Mass budget is a preliminary mechanical allocation; final value requires enclosure/PCB CAD.
A9	4/12/16-element SWaP values are first-order linear scaling from the 8-element reference design.
A10	Patch-element dimensions are unspecified; 35 cm outer-diameter compliance must be confirmed with the selected patch footprint.
A11	1621 MHz satphone and 1240 MHz radar EIRP/range are unspecified, so their effects are qualitative.

References

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- [14] SiTime, SiT9366 Ultra-Low-Jitter Differential Oscillator Data Sheet.
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